Table 7. Circuit validation

Panel A. Validation metrics

Metric	Validation layer	Value	Tolerance	Passes	Qiskit execution status	Status
numpy_embedding_statevector_error	NumPy embedding	5.551e-17	1.000e-12	Yes	executed	qiskit executed
numpy_embedding_doublet_observable_error	NumPy embedding	0	1.000e-12	Yes	executed	qiskit executed
numpy_embedding_quartet_observable_error	NumPy embedding	0	1.000e-12	Yes	executed	qiskit executed
numpy_embedding_leakage_probability	NumPy embedding	0	1.000e-24	Yes	executed	qiskit executed
qiskit_statevector_error	Qiskit	5.551e-17	1.000e-12	Yes	executed	qiskit executed
qiskit_doublet_observable_error	Qiskit	0	1.000e-12	Yes	executed	qiskit executed
qiskit_quartet_observable_error	Qiskit	0	1.000e-12	Yes	executed	qiskit executed
qiskit_leakage_probability	Qiskit	0	1.000e-24	Yes	executed	qiskit executed
qiskit_basis_ordering_error	Qiskit	0	1.000e-15	Yes	executed	qiskit executed

Circuit checks cover coherent statevector execution and six state to eight state embedding only. Presentation values are rounded. All 9 source rows and all 29 source columns remain in table07_circuit_validation.csv; 9 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 7. Circuit validation

Panel B. Executed coherent calculation

Metric	Seed	Qubits	Basis order	Physical indices	Duration (s; illustrative conversion)	Input state origin	State preparation	Implementation type
numpy_embedding_statevector_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
numpy_embedding_doublet_observable_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
numpy_embedding_quartet_observable_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
numpy_embedding_leakage_probability	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
qiskit_statevector_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
qiskit_doublet_observable_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates

Circuit checks cover coherent statevector execution and six state to eight state embedding only. Presentation values are rounded. All 9 source rows and all 29 source columns remain in table07_circuit_validation.csv; 9 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 7. Circuit validation

Panel B. Executed coherent calculation (continued)

Metric	Seed	Qubits	Basis order	Physical indices	Duration (s; illustrative conversion)	Input state origin	State preparation	Implementation type
qiskit_quartet_observable_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
qiskit_leakage_probability	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates
qiskit_basis_ordering_error	1729	3	["000", "001", "010", "100", "101", "110"]	[0, 1, 2, 4, 5, 6]	1.000e-06	normalized pseudorandom six-element complex statevector supplied directly with seed 1729	none; Statevector input is supplied numerically	one inserted dense 8x8 UnitaryGate; not decomposed gates

Circuit checks cover coherent statevector execution and six state to eight state embedding only. Presentation values are rounded. All 9 source rows and all 29 source columns remain in table07_circuit_validation.csv; 9 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 7. Circuit validation

Panel C. Observables and exclusions

Metric	Measurement	Post processing	Scope	Reason	Does not validate
numpy_embedding_statevector_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
numpy_embedding_doublet_observable_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
numpy_embedding_quartet_observable_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
numpy_embedding_leakage_probability	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
qiskit_statevector_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage

Circuit checks cover coherent statevector execution and six state to eight state embedding only. Presentation values are rounded. All 9 source rows and all 29 source columns remain in table07_circuit_validation.csv; 9 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 7. Circuit validation

Panel C. Observables and exclusions (continued)

Metric	Measurement	Post processing	Scope	Reason	Does not validate
qiskit_doublet_observable_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
qiskit_quartet_observable_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
qiskit_leakage_probability	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage
qiskit_basis_ordering_error	none; no measurement gates or finite-shot sampling	numerical statevector difference, D/Q projector expectations, basis ordering, and unused-state leakage	closed coherent simulator and encoding consistency only	Not available	independent Hamiltonian construction; spin-selective reaction; escape; Lindblad relaxation; upstream or downstream chemistry; hardware performance; quantum advantage

Circuit checks cover coherent statevector execution and six state to eight state embedding only. Presentation values are rounded. All 9 source rows and all 29 source columns remain in table07_circuit_validation.csv; 9 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.